

Class Project Phase III – Part 1 Submission Template

Group 6

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Explain your choice for the software architecture and comparison with other alternatives (at least one). This explanation should include benefits of using the software architecture of your choice. Why is it better than the other.

Make sure to include your architecture diagram using C4 model in your explanation. Your architecture diagram should have two sub diagrams: the context diagram and the component diagram. Ten points extra credit can be earned if you include the deployment diagram in addition to the context diagram and the component diagram. Module 10 has examples of various architecture diagrams and the C4 model examples.

Length: This can be 2-3 pages long including the architecture diagram (12 font, single page document).

Point allocation: Explanation 7.5 points, Architecture Diagram 7.5 points.

-----Start Your Document Below-----

Architecture explanation and architecture diagram [15 Points]:

Architecture Explanation:

For the SunDevil Connect project, we selected a Layered Architecture because it provides a clear separation of responsibilities and supports future system growth. The application contains several features including user authentication, event management, club management, membership tracking, announcements, and administrative moderation. Separating these responsibilities into layers makes the system easier to understand, maintain, and expand.

The first layer is the Presentation Layer, which contains the user interface pages such as LoginPage, EventSearchPage, EventDetailsPage, ClubPage, ClubDashboardPage, and AdminPanelPage. This layer is responsible for displaying information and collecting user input.

The second layer is the Business Logic Layer. This layer contains controllers such as AuthController, EventController, ClubController, and AdminController. These controllers process requests from users and enforce business rules. The AuthenticationManager singleton identified during Phase II is also located in this layer.

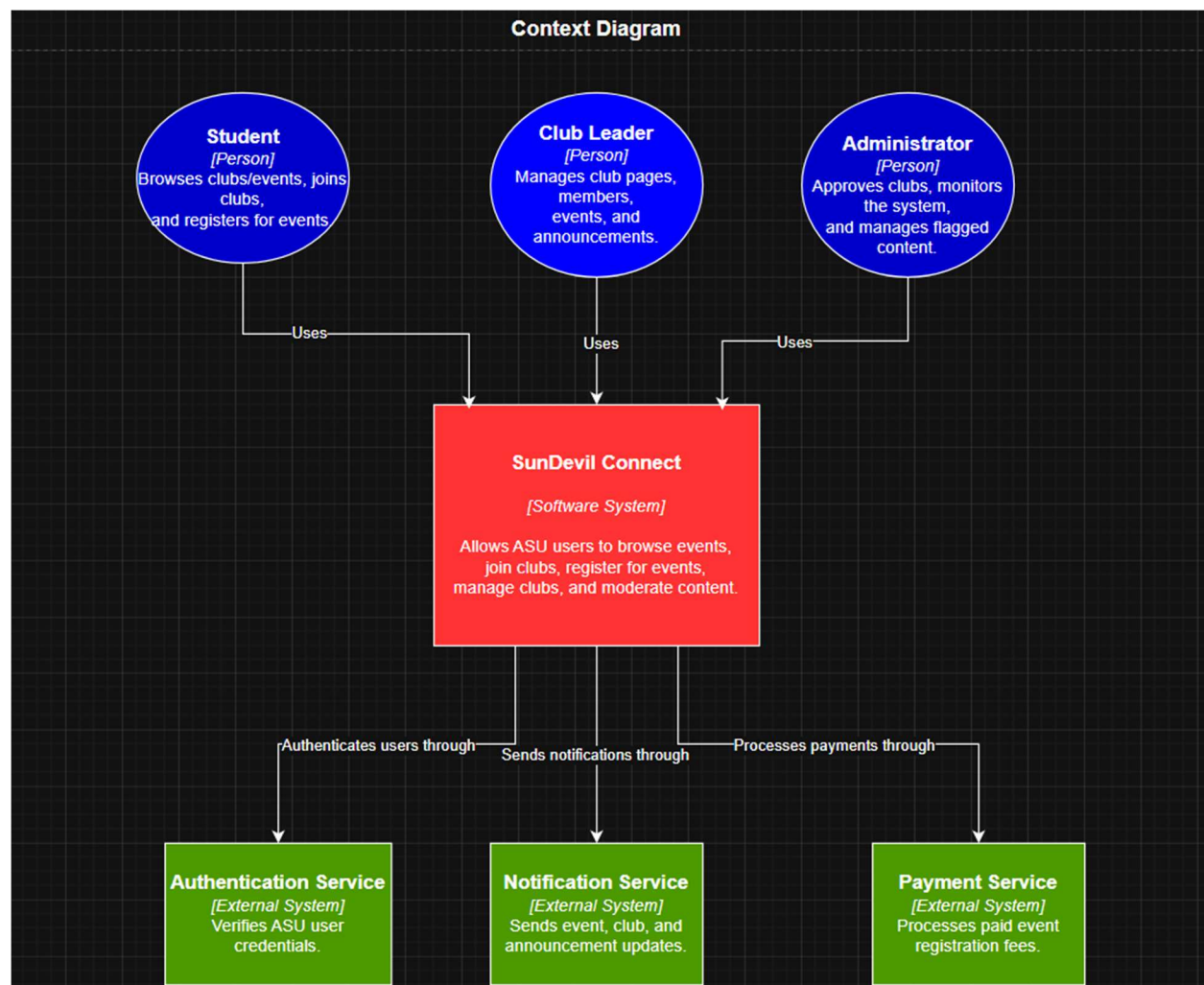
The third layer is the Data Layer. This layer contains the entity classes including User, Student, ClubLeader, Administrator, Club, Event, EventRegistration, ClubMembership, Announcement, and

FlaggedContent. These classes represent the data used by the system and store information required for application functionality.

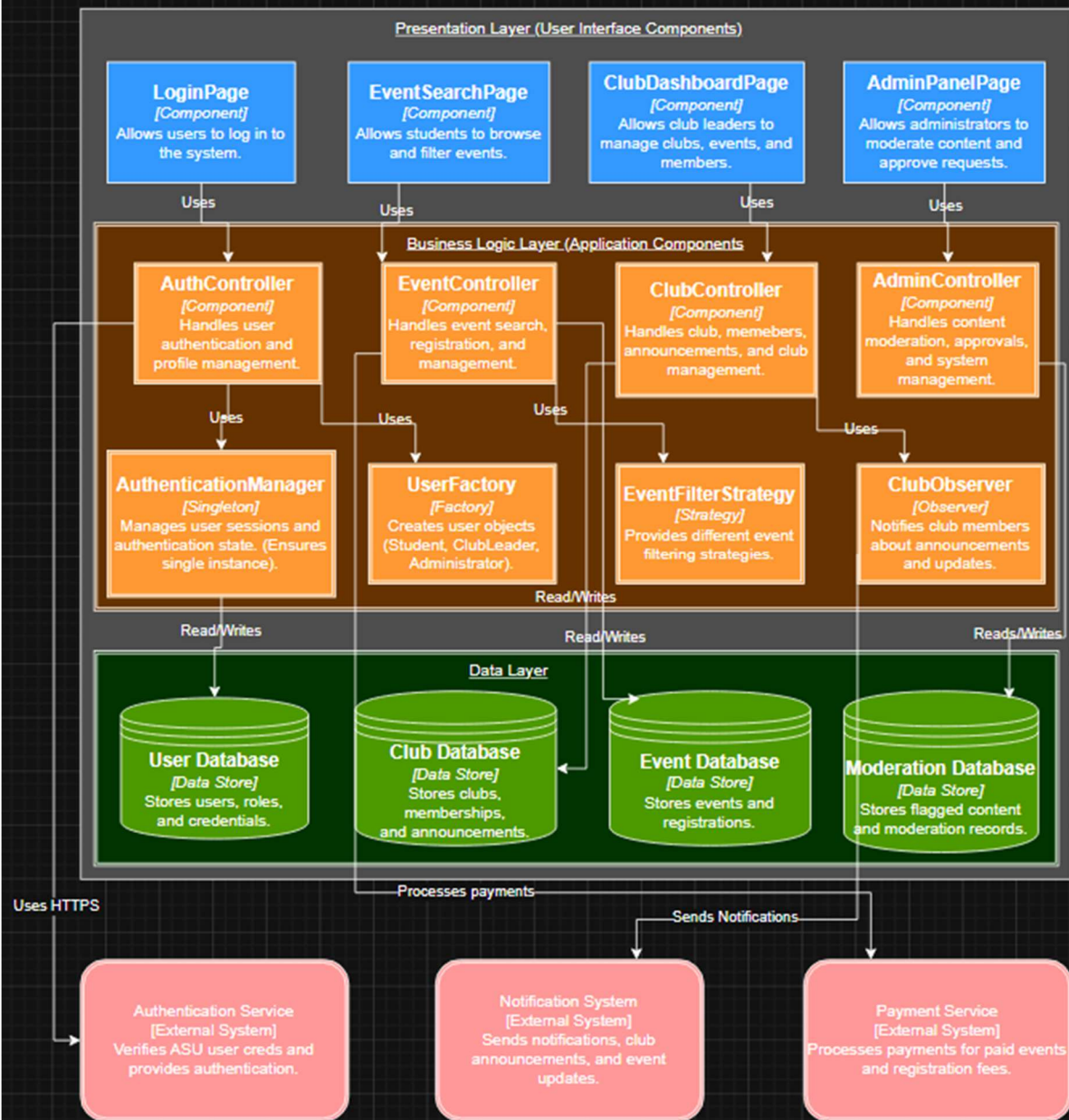
We selected Layered Architecture because it improves maintainability and scalability. Changes to the user interface can be made without affecting data storage, while updates to business rules can be implemented without modifying the presentation layer.

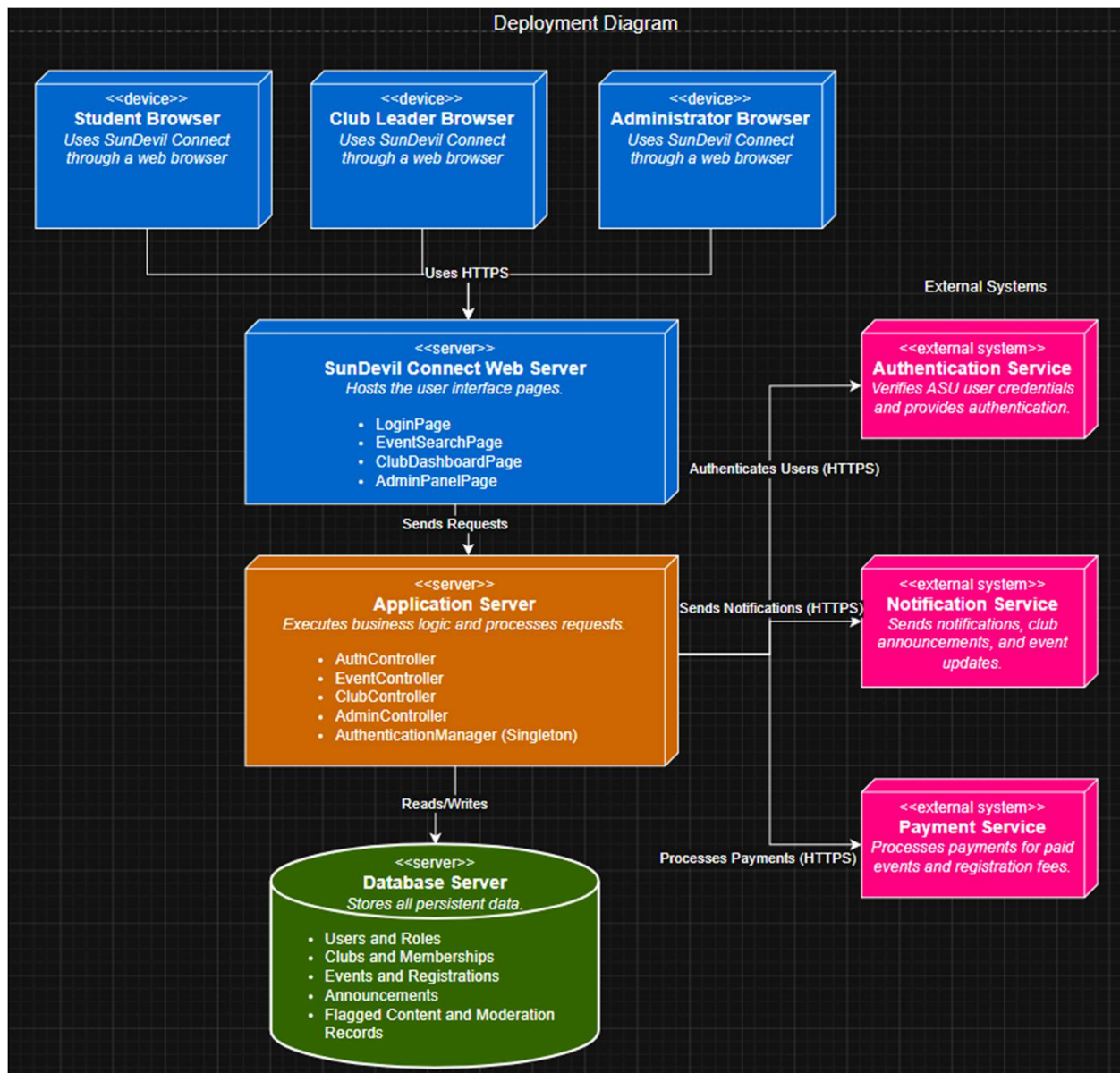
An alternative architecture considered was a Monolithic Architecture. While a monolithic approach may be easier to develop initially, all functionality is tightly coupled within a single application. As the system grows, maintenance becomes more difficult and changes can affect multiple parts of the application. Layered Architecture provides better separation of concerns, improved code organization, and easier future expansion. Because SunDevil Connect is expected to support many users, clubs, and events, Layered Architecture is a better choice for long-term development and maintenance.

Architecture Diagrams:



Component Diagram





Phase I code [10]: Please upload the code you have developed for this phase into [the google drive and copy and paste the link.

https://drive.google.com/drive/folders/1OY7eZljLP8UapehsmDTOvwgJwDxzPAR?usp=drive_link

Video Link [5]: Prepare a 3-5 minute video explaining your implementation and dome phase I implementation. Upload it to YouTube or any online repository and provide the link.

https://drive.google.com/file/d/1OUD9nnLvxCV-cn-1XQxKaIBkTQvN56ud/view?usp=drive_link